Toowoomba's quiet week: what changed when its repeater went off air

Investigation 02 · 29 October 2026 · Toowoomba AL (540162, rain 6598), repeater on 151.525 MHz, Brisbane hub · Open: Toowoomba implicated, site fault not yet found

- **Toowoomba's transmitter last spoke at 04:35 on 25 October.** Its rain counter had reset to 0 at 03:16, which looks like a power cut or a restart. Its carrier hasn't been heard since.
- **On its channel, corrupted copies fell to a third at that moment**: from 53 to 19 per 1,000 decoded, with the same traffic, back where it was in early October. Our receiver's update the day before didn't do it: the rate stayed at 46 until Toowoomba stopped.
- **Toowoomba was sending some of the damage itself.** 8.4% of the bursts on its carrier were corrupted copies, about twice the share on the other two repeaters on its channel. Mt Glorious, which relays the same addresses, went from 3.8% to 2.0% once Toowoomba stopped.
- **It looks to have been failing since 15 October.** That day the corruption on its channel tripled, and its own three-hourly reports began going missing.

The test: the channel cleared when Toowoomba stopped

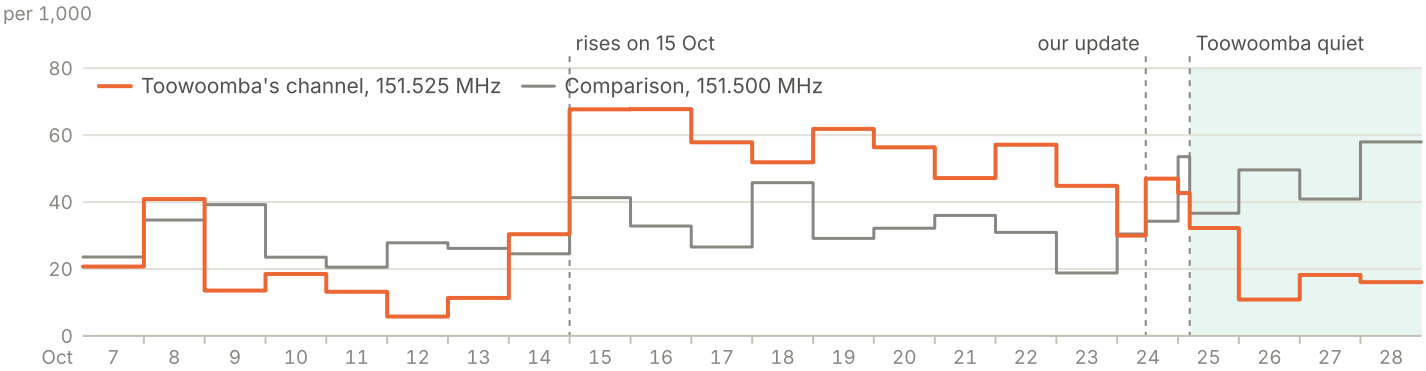


Figure 1. Corrupted copies per 1,000 decoded, each day. Orange: Toowoomba's channel. Grey: 151.500 MHz, our receiver's only other channel all fortnight, which Toowoomba doesn't use. Green: Toowoomba's silence. Hover a day for its counts.

On Toowoomba's channel the rate ran at 19 per 1,000 in early October and rose to 58 from 15 October, with no change in traffic (about 1,018 and 1,119 bursts a day). When Toowoomba went quiet it dropped to 19, still on about 1,126 bursts a day: 82 corrupted copies in 4,289, against 921 in 17,361 from 15 October up to the silence. That's 0.36 times the earlier rate (95% range 0.29–0.45), too large a drop for chance. The comparison channel didn't drop at 04:35, so the fall wasn't something at our end. 98% of the corrupted copies on Toowoomba's channel were clear: as strong as the clean copy, so wrong when they were sent, not misheard.

What went quiet at 04:35 on 25 October

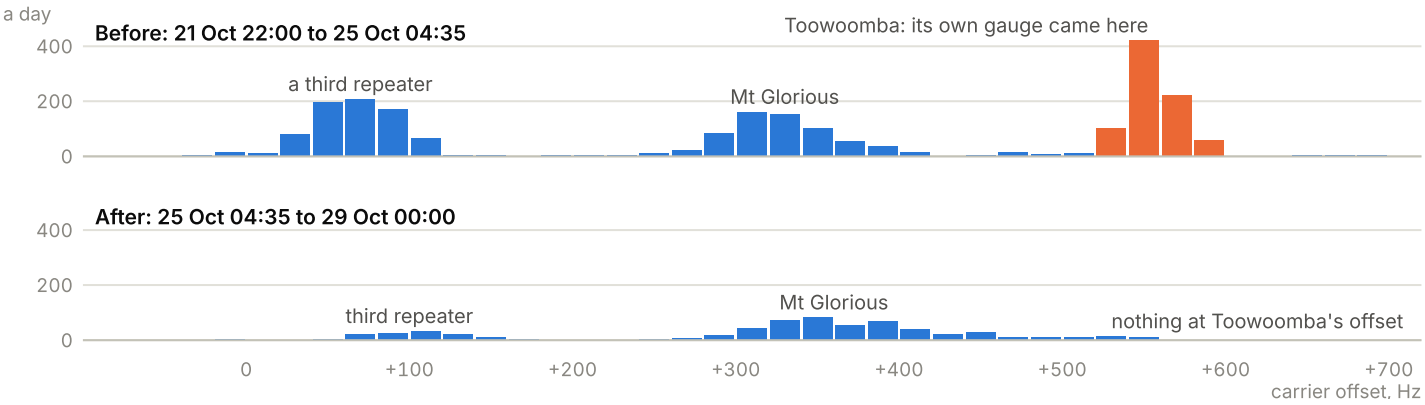


Figure 2. Bursts a day on 151.525 MHz by carrier offset, in 20 Hz steps, since offsets were first recorded (3.3 days before, 3.8 after). Orange is Toowoomba's carrier. Every carrier reads about 40 Hz higher after 24 October, so look at the shape, not the exact offset.

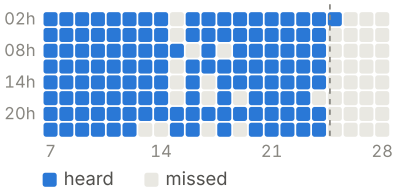


Figure 3. Toowoomba's three-hourly reports, 7–28 October. The dashed line is 04:35 on 25 October.

The tall peak near +560 Hz is Toowoomba. 91% of its 2,638 bursts were addresses in Toowoomba's pass ranges, and all 128 reports from its own gauge came on it. After 04:35 on 25 October, one burst from its ranges arrived anywhere near that offset. The Mt Glorious peak carries addresses only Mt Glorious relays, and it's still there.

Before 15 October almost every one of its three-hourly reports arrived (2 of 64 missed). On the 15th, the day the corruption rose, 5 of 8 went missing, and 10 by the 18th. On 25 October its count read 222, then 0 at 03:16, then 1 to 4 by 04:23 as it rained. Its carrier was last heard at 04:35, 78 minutes after the reset. The one copy of its address since (27 October, 18:02) came on Mt Glorious's carrier with a count of 655: another gauge, not Toowoomba.

Before and after, side by side

Each window is a stretch of the fortnight. The orange bars stay high through the storm and through our receiver's update, 17 hours before the silence (69 corrupted in 1,499, 46 per 1,000). They drop only when Toowoomba stops. The grey bars, on the comparison channel, don't drop then: 39 per 1,000 just before, 47 after.

Toowoomba's own addresses, which now reach us only through the repeaters that share its ranges, went from 87 to 38 per 1,000 (32 in early October). Addresses outside its ranges went from 26 to 10 (8 early), so it was spoiling traffic it doesn't list as well.

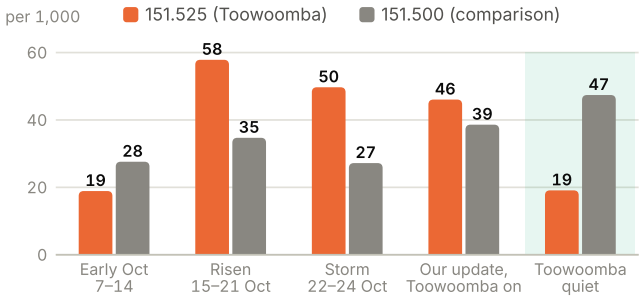


Figure 4. Corrupted copies per 1,000 decoded in each window. The shaded pair is Toowoomba's silence. Hover a bar for its counts.

Where the damage came from

From 21 October, when carriers were first recorded, to the silence, 222 corrupted copies came on Toowoomba's carrier: 54% of all those on the channel, and 8.4% of what that carrier brought. The other two repeaters on the channel ran at 3.8% (Mt Glorious) and 4.2% (the third). Once Toowoomba stopped, Mt Glorious's share fell to 2.0%. It relays Toowoomba's addresses onward, so some of its damage was Toowoomba's, passed on. The third repeater stayed about where it was (5.0%).

Flood-Net's transmitter table, as judged at 05:13 on 29 October, puts 85 corrupted copies on Toowoomba against 327 bursts (26%), against a typical 0.39%. That overstates it. The table divides by the bursts it pinned to Toowoomba itself, not everything on its carrier, and it gave most of Toowoomba's carrier to Mt Glorious, which relays the same addresses. Measured on the whole carrier, Toowoomba's share is the 8.4% above.

What we know

Established. Toowoomba's transmitter stopped at 04:35 on 25 October. On its channel, corrupted copies fell from 53 to 19 per 1,000 at that moment, on the same traffic. Neither our receiver's update nor anything else at our end caused it: the comparison channel didn't drop.

Likely. Toowoomba was failing for ten days before it died, and sending corrupted copies the whole time. Some went out again through Mt Glorious. A counter reset, then silence 78 minutes later in rain, points at power: battery, charger or supply.

Open. What failed on 15 October. Why traffic outside Toowoomba's ranges was damaged too: it may relay more than its pass ranges say, or its transmissions may have been colliding with other repeaters' bursts.

Next

1. Visit Toowoomba AL: power first (battery, charger, solar, supply), then the radio. Ask whether anything was done at the site around 15 October.
2. When it's back on air, watch Figure 1's orange line. If it stays near 19 per 1,000, the fault went with the repair; if it climbs back towards 53, it didn't.
3. Fix the transmitter table so it uses the whole carrier as the denominator, and so a repeater can't be credited with another's carrier. The Bit-Flip Sources tab and the bulletins read from it.

Method: Flood-Net's receiver, legacy ALERT bursts on 151.525 MHz (Toowoomba's channel) and 151.500 MHz (the comparison), 7–28 October (Brisbane time). A corrupted copy is one Flood-Net's checks found: a burst one or two bits from a real one, on the same receiver within five seconds. Rates are corrupted copies over bursts decoded. An address is Toowoomba's if it is in Toowoomba AL's pass ranges in the station register. The receiver's other channels changed during the fortnight, so they are left out. Data, queries and script: `data/investigations/2026-10-toowoomba-quiet-week/`, `tools/investigations/toowoomba_quiet_week.sql`, `tools/investigations/toowoomba_quiet_week.py`.

Words used here.

Corrupted copy	A second copy of a real report, heard within five seconds of it, with one or two of its address or value bits changed.
Per 1,000 decoded	Corrupted copies for every 1,000 bursts our receiver decoded on that channel, all addresses together.
Carrier offset	How far a transmitter's carrier sits from the channel's centre, as our receiver measures it. Every crystal is off by its own amount, so the offset tells apart transmitters that share a channel.
Pass range	The block of addresses a repeater is set to relay. Toowoomba passes 4000–4099, 6526–6527 and its own gauge, 6598.
Clear	A corrupted copy as strong as the clean one, within 6 dB: the transmitter sent wrong bits, rather than our receiver mishearing a weak one.
Three-hourly report	Toowoomba's gauge sends its count at two minutes past 2, 5, 8, 11, 14, 17, 20 and 23 o'clock, rain or not.